

QOI & LQOI Technical Reference

Part I: Original QOI (Quite OK Image) Format

QOI is a greedy, single-pass, lossless image compression algorithm. It operates row-major and maintains minimal state, achieving extreme encode/decode speeds on single-core CPUs.

Global State:

1. **Previous Pixel (px_prev):** Initialized to r:0, g:0, b:0, a:255.
2. **Hash Array (index[64]):** A 64-element array of previously seen pixels, zero-initialized.
 - **Hash Function:** $(R \times 3 + G \times 5 + B \times 7 + A \times 11) \bmod 64$

File Structure:

- **Header (14 bytes):** Magic qoif (4), Width (4), Height (4), Channels (1), Colorspace (1).
- **Data Stream:** Sequence of variable-length, byte-aligned chunks.
- **End Marker (8 bytes):** 0x00 (x7) followed by 0x01.

Encoding Methods (Opcodes):

- **QOI_OP_RUN (11xxxxxx):**
 - *Payload:* 6-bit run length (1 to 62), biased by -1.
 - *Trigger:* Current pixel exactly matches px_prev.
- **QOI_OP_INDEX (00xxxxxx):**
 - *Payload:* 6-bit index (0 to 63).
 - *Trigger:* Current pixel exactly matches index[hash].
- **QOI_OP_DIFF (01xxxxxx):**
 - *Payload:* 2-bit ΔR , ΔG , ΔB .
 - *Trigger:* Δ for RGB are each within $[-2, 1]$. Stored with +2 bias.
- **QOI_OP_LUMA (10xxxxxx):**
 - *Payload:* Byte 1: 6-bit ΔG $[-32, 31]$. Byte 2: 4-bit $\Delta R - \Delta G$ $[-8, 7]$ and 4-bit $\Delta B - \Delta G$ $[-8, 7]$.
 - *Trigger:* ΔG and relative ΔR , ΔB fall within respective ranges. Stored with +32 and +8 biases.
- **QOI_OP_RGB (11111110) / QOI_OP_RGBA (11111111):**
 - *Payload:* 3 or 4 full bytes of raw channel data.
 - *Trigger:* Fallback when no other opcode applies.

Part II: Lossy QOI (LQOI) Implementation Specification

LQOI introduces perceptual quantization directly into the encoding logic. The following specifications detail the exact mathematical and state changes required to implement the lossy variant.

1. Green-Weighted Manhattan Distance (Global Error Function)

- **Mechanism:** Replaces exact-match checks with a branchless L1 norm weighted heavily toward the Luma (Green) channel.

- **Formula:** $E = 2 \times |\Delta G| + |\Delta R| + |\Delta B|$
- **Threshold:** $E \leq 6$ (Alpha channel must always exactly match; $\Delta A = 0$).

2. Chroma-Biased Lossy Runs

- **Mechanism:** A run continues as long as the current source pixel's error E relative to the *initiating base pixel* of the run is ≤ 6 .
- **Implementation:**
 - Store $px_base = current_px$ when a run begins.
 - For subsequent pixels, if $(2 \times |px.g - px_base.g| + |px.r - px_base.r| + |px.b - px_base.b|) \leq 6$ AND $px.a == px_base.a$, increment run.
- **Time:** Decreased (bypasses hash/diff math).
- **Compression:** Massively increased.
- **Error:** Strictly bounded by $E \leq 6$ relative to the start of the run, preventing compounding drift.

3. Lossy Indexing (Snap to Palette)

- **Mechanism:** Allows a pixel to "snap" to a perceptually similar color already residing in the hash array by using a **locality-sensitive hash**. This ensures similar colors collide into the same bucket.
- **Implementation:**
 - Calculate $H = ((R \& 0xF8) \times 3 + (G \& 0xF8) \times 5 + (B \& 0xF8) \times 7 + A \times 11) \bmod 64$. (*Masking the bottom 3 bits of RGB ensures perceptually similar colors yield the same hash bucket*).
 - Let $pal_px = index[H]$.
 - If $px.a == pal_px.a$ AND the Green-Weighted Manhattan Distance between px and pal_px is ≤ 6 , emit QOI_OP_INDEX with payload H .
 - *Crucial:* Set $px = pal_px$ for all subsequent encoding steps (Base Pixel Hash Injection).
- **Time:** Neutral (+3 cycles for integer bounds check).
- **Compression:** Massively increased.
- **Error:** Bounded by $E \leq 6$.

4. Base Pixel Hash Injection

- **Mechanism:** The hash array and px_prev state must only track the *quantized/substituted* values, not the raw source values.
- **Implementation:** When a pixel is absorbed into a Lossy Run or Lossy Index, the encoder updates $index[H]$ and px_prev using the base/palette pixel's RGB values.
- **Time:** Neutral.
- **Compression:** Increased (standardizes the hash array).
- **Error:** Eliminates decoder mismatch; the decoder only operates on exact bitstreams.

5. Repurposed Diff Ranges (QOI_OP_DIFF)

- **Mechanism:** Because values $[-1, 1]$ are absorbed by Lossy Runs and Indexing, QOI_OP_DIFF is permanently shifted to handle wider disconnected ranges.
- **Implementation:**
 - Valid raw differences: $\{-4, -3, 2, 3\}$.
 - 2-bit mapping (Encoder):
 - $00 \rightarrow -4$
 - $01 \rightarrow -3$
 - $10 \rightarrow 2$
 - $11 \rightarrow 3$

- Bitwise implementation: $\text{encoded_val} = (\text{raw_diff} < 0) ? (\text{raw_diff} + 4) : (\text{raw_diff})$.
- **Time:** Neutral.
- **Compression:** Increased (captures medium jumps in 1 byte).
- **Error:** Introduces quantization since intermediate values (e.g., $\Delta = -2$) must be snapped to the nearest valid bin.

6. Scaled QOI_OP_LUMA

- **Mechanism:** Bit-shifts the Green channel difference payload to double its reach, capturing high-contrast edges in 2 bytes instead of 4.
- **Implementation:**
 - **Encoder:**
 - $\text{actual_dg} = \text{px.g} - \text{px_prev.g}$
 - $\text{encoded_dg} = \text{actual_dg} \gg 1$ (Bitwise arithmetic right shift).
 - Range of encoded_dg must fall in $[-32, 31]$, making the effective capture range of actual_dg $[-64, 63]$. Store with +32 bias.
 - dr_dg and db_dg are calculated relative to the *decoded* green difference to prevent cascading error: $\text{decoded_dg} = \text{encoded_dg} \ll 1$.
 - **Decoder:**
 - $\text{dg} = (\text{payload} - 32) \ll 1$.
- **Time:** Neutral (single-cycle bitshift).
- **Compression:** Greatly increased.
- **Error:** Introduces a ± 1 quantization error exclusively on the Green channel for large color jumps.